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# AI for Transportation Planning & Design (AI TPD)

*Session Name: AI for Transportation Planning & Development*  
*Date: August 20, 2026*



U.S. Department of Transportation  
ITS Joint Program Office

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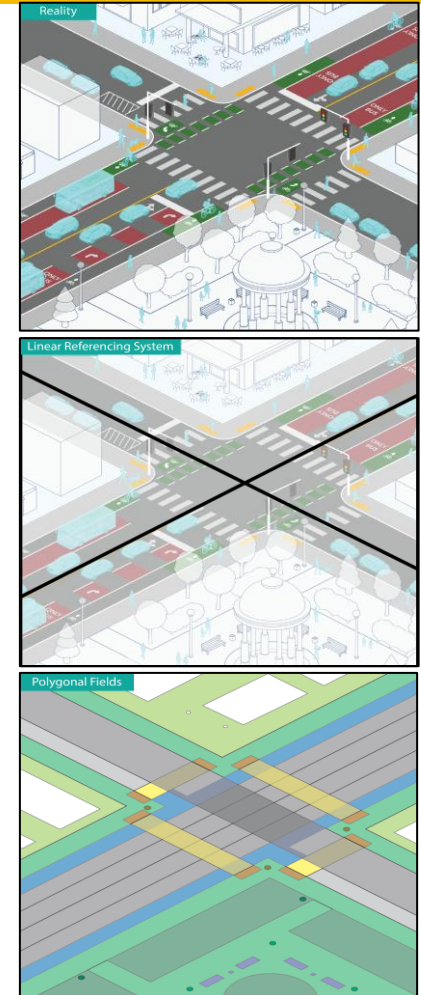
# AI TPD Initiative: Challenge & Gaps

## Challenge

- Multimodal data collection and integration can be slow, expensive, and quickly outdated, especially for active transportation.
  - *Example: Manual sidewalk mapping costs ~\$100 to \$800 per mile.*

## Gap

- Agencies lack user-friendly tools to easily identify multimodal system needs, use, condition and safety risks along with responsive strategies and projects.



# AI TPD Initiative: Overview & Objectives

**Core Goal:** Develop AI-powered planning software that uses high-resolution data to streamline systems planning (trends analysis, needs identification & high-level strategies), project design & early project development.

## Critical Questions Answered

- *Planning:* Where are near-misses occurring before they turn into severe or fatal crashes?
- *Design:* What specific safety and infrastructure upgrades will work best at this location?

# AI TPD Initiative: Progress

## **Phase I: Proof of Concept (Completed)**

Successfully demonstrated new AI methods to automatically extract and map multimodal data in initial test areas.

## **Phase II: Working Software (Underway)**

Scaling data mapping across larger geographies to build fully functioning, interactive planning and design decision platforms.

# AI TPD Initiative: Initial Lessons Learned

- **Data Source Flexibility:** Aerial imagery quickly maps large networks, but street-level views or laser scans are necessary to capture precise details like curb heights and slopes.
- **Weather Adaptability:** Severe weather (such as snow) can block camera views, meaning AI models must combine multiple data sources to remain reliable.

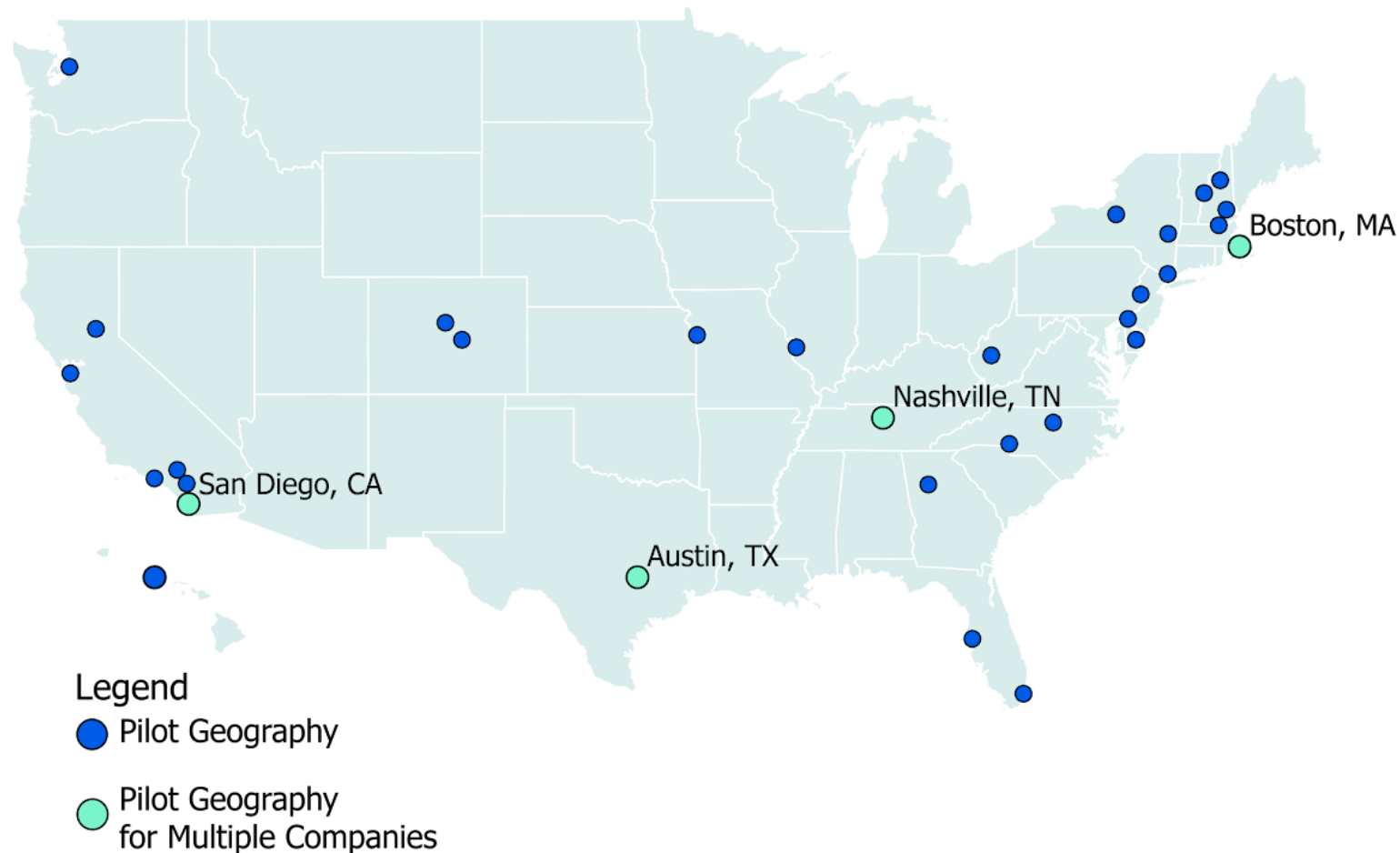
# AI TPD Initiative: Initial Lessons Learned

- **Integrated Software is Key:** Transportation agencies do not just want raw data layers; they need software that integrates with existing processes on planning, programming (funding), and early design decisions.
- **Collaborative Development:** Partnering directly with local practitioners is critical to ensuring AI tools solve real-world, data and workflow challenges.

# AI TPD Initiative: Current Technical Work

- **Scaling to Additional Test Environments:** Evaluating and validating data and AI methods across multiple new geographies to ensure reliability.
- **Ensuring Broad Usability:** Testing will cover urban, suburban, and rural communities, multimodal networks, and mix of system conditions.
- **Continuing Intensive Collaboration:** AI solution is being developed in close partnership with local agencies to ensure outputs integrate seamlessly into existing data systems, processes & workflows.

# AI TPD Initiative: Current Technical Work



**Each  
company  
creating and  
testing  
products  
with 5 new  
geographies**

# AI TPD Initiative: Intended Results

- **Predictive Multimodal Safety Needs:** Use local video and near-miss sensor data to identify critical safety gaps and needs.
- **Reduce Analysis Time:** Demonstrate how decision-support tools can enhance planning analysis and project prioritization times using AI-assisted methods.
- **Automated Digital Twins:** Create highly detailed, 3D virtual models of local streets directly from standard, low-cost street images.

# Thank You



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